

Professional Computing Practice BITM 402

Instructions for Mid-Semester Ethical Debate Assignment

Debate Session:

- The debate will be held in Week 8 during class hours.
 - The total possible score for this debate is 10 marks.
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Instructions to Students:

1. Group Assignment:

Each scenario has two clearly defined groups: one group arguing **FOR** and the other **AGAINST** the ethical question presented. Ensure you clearly understand your assigned scenario and your group's position.

2. Preparation Requirements:

- Develop coherent, logical arguments clearly supporting your group's ethical stance.
- Base your arguments on ethical principles, theories, and frameworks discussed in class.
- Conduct additional research, referencing credible academic sources, real-world examples, and relevant ethical standards.
- Anticipate possible counter-arguments from the opposing side and prepare clear, reasoned rebuttals.

3. Debate Format:

Each group will:

- Present their main argument (5 minutes maximum per group).
- Engage in direct rebuttals and responses to opposing group (additional 5 minutes per scenario).
- Answer follow-up questions posed by classmates or instructor (open Q&A, 5 minutes maximum).

4. Critical Thinking and Ethical Reasoning:

Regardless of personal beliefs, each group must thoroughly explore and articulate

ethical implications from the assigned perspective. Demonstrate empathy, objectivity, and a nuanced understanding of complex ethical dilemmas.

Assessment Rubric (Total: 10 Marks)

Criteria	Marks
Clarity and Structure	2 Marks- Arguments are logical, clearly structured, and easy to follow.
Use of Ethical Principles/Theories	3 Marks- Clear application of ethical theories and principles relevant to the scenario.
Quality of Evidence and Examples	2 Marks- Arguments supported by relevant, credible sources, real-world examples, or case studies.
Engagement and Rebuttals	2 Marks- Effectively addresses opposing arguments, showing deep understanding and insightful responses.
Team Participation and Collaboration	1 Mark- Demonstrates equal participation, teamwork, and coordination within the group.

Scenario 1: Security Vulnerability & Responsible Disclosure

Case Study:

A software engineer, Harper, discovers a critical security flaw in a competitor's operating system. Exploiting it could give Harper's company a competitive edge, but it poses a risk to millions of users if left unpatched.

Debate Question:

Is it ethical for Harper to publicly disclose the security flaw immediately—even if it risks causing widespread panic or potential exploits—rather than privately contacting the competitor and coordinating a patch?

- **Group 1 (FOR):** Argue that immediate public disclosure is ethically necessary to alert users quickly.
- **Group 2 (AGAINST):** Argue that private, responsible disclosure is ethically required to prevent harm.

Scenario 2: AI Monitoring in the Workplace

Case Study:

An AI startup offers a productivity monitoring system that tracks remote employees' keystrokes, mouse movements, and webcam footage. Management claims productivity increases, but employees feel it's invasive.

Debate Question:

Is it ethically justifiable for employers to implement intrusive AI monitoring to track their remote staff's every digital and physical move?

- **Group 3 (FOR):** Argue that intrusive AI monitoring is ethically justified for protecting company productivity and resources.
 - **Group 4 (AGAINST):** Argue that such invasive monitoring violates employees' privacy and trust, making it unethical.
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Scenario 3: Social Media & Data Sharing

Case Study:

A social media platform collects detailed personal data and sells it to third-party advertisers for targeted marketing. Data usage practices are buried in lengthy terms of service.

Debate Question:

Is it ethical for the social media platform to sell detailed user data to advertisers, given that most users do not fully understand the data practices?

- **Group 5 (FOR):** Argue that selling user data ethically funds free and personalized services.
 - **Group 6 (AGAINST):** Argue it is unethical due to users' lack of informed consent and transparency.
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Scenario 4: Biased Algorithmic Hiring

Case Study:

A large tech firm's AI hiring system disproportionately rejects women and older applicants. Developers claim bias was unintentional.

Debate Question:

Is it ethical for companies to continue using an AI-driven hiring system once they become aware of potential biases, especially if it improves efficiency and reduces overhead costs?

- **Group 7 (FOR):** Argue that continued use is ethically permissible if it significantly boosts efficiency, and biases can be gradually corrected.
 - **Group 8 (AGAINST):** Argue it's unethical due to inherent discrimination that violates fairness and equality principles.
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Scenario 5: Personalizing Education with Data Analytics

Case Study:

An educational software provider collects data on students to tailor lessons and sells anonymized student data to research institutions.

Debate Question:

Is it ethical for the educational software company to collect and share detailed student performance data—even if it's anonymized—for the sake of refining personalized education and securing funding?

- **Group 9 (FOR):** Argue that data sharing is ethical, improving education and innovation through anonymized insights.
 - **Group 10 (AGAINST):** Argue it's unethical due to privacy risks and potential misuse of student data without explicit consent.
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Scenario 6: Autonomous Vehicles and Moral Dilemmas

Case Study:

An autonomous vehicle manufacturer prioritizes occupant safety over pedestrians during accidents.

Debate Question:

Is it ethically permissible for autonomous vehicle companies to prioritize occupants' lives over pedestrians in life-threatening scenarios?

- **Group 11 (FOR):** Argue prioritizing occupant safety is ethical as they are the direct consumers of the product.
 - **Group 12 (AGAINST):** Argue pedestrian safety must ethically hold equal importance, making occupant prioritization unethical.
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Scenario 7: Genetic Editing in Humans

Case Study:

A biotechnology firm develops gene-editing technology to eliminate genetic diseases but sparks concerns about “designer babies.”

Debate Question:

Is it ethical to allow gene-editing in human embryos, even if initially intended only to prevent serious genetic diseases?

- **Group 13 (FOR):** Argue gene-editing ethically prevents human suffering and enhances health.
 - **Group 14 (AGAINST):** Argue it is unethical due to risks of inequality and misuse in creating “designer babies.”
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Scenario 8: Surveillance Drones and Public Safety

Case Study:

A city government deploys facial recognition-equipped surveillance drones to reduce crime in public spaces, sparking privacy concerns.

Debate Question:

Is it ethical for governments to use surveillance drones with facial recognition technology in public spaces to maintain public safety?

- **Group 15 (FOR):** Argue drone surveillance is ethically justified for public safety and crime prevention.
 - **Group 16 (AGAINST):** Argue it is unethical due to significant privacy infringements and potential misuse.
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Scenario 9: Smart Home Devices and Data Privacy

Case Study:

A smart home company extensively collects private home data to improve device functionality and predictive analytics.

Debate Question:

Is it ethical for smart home device manufacturers to extensively collect and analyze private home activity data to enhance product capabilities?

- **Group 17 (FOR):** Argue extensive data collection is ethical for improved user convenience and device performance.
 - **Group 18 (AGAINST):** Argue extensive data collection is unethical as it severely violates consumer privacy and autonomy.
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Scenario 10: Environmental Impact vs. Technological Advancement

Case Study:

A corporation builds a large data center in a rural area, promising employment and connectivity but causing significant environmental harm.

Debate Question:

Is it ethical for corporations to pursue significant technological advancement that creates local employment at the expense of substantial environmental degradation?

- **Group 19 (FOR):** Argue that technological advancement and local economic growth ethically justify environmental harm.
 - **Group 20 (AGAINST):** Argue that such environmental degradation is unethical and unjustifiable regardless of technological or economic benefits.
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Groups and Assigned Topic Summary Table:

Scenario	Ethical Debate Topic	FOR	AGAINST
1	Security Vulnerability Disclosure	Group 1	Group 2
2	AI Monitoring in Workplace	Group 3	Group 4
3	Social Media Data Sharing	Group 5	Group 6
4	Biased Algorithmic Hiring	Group 7	Group 8
5	Personalizing Education Data Analytics	Group 9	Group 10
6	Autonomous Vehicle Moral Dilemma	Group 11	Group 12
7	Genetic Editing in Humans	Group 13	Group 14
8	Surveillance Drones and Public Safety	Group 15	Group 16
9	Smart Home Devices Data Privacy	Group 17	Group 18
10	Environmental Impact vs. Technological Advancement	Group 19	Group 20